Michael Cogan

Embedded Software Engineer

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SUMMARY

Embedded software engineer with hands-on experience across firmware-level C/C++, Linux systems, and real sensor/peripheral integration. Built an autonomous AI system running locally on server hardware with a Raspberry Pi sensing node, plus production web backends. A clear communicator and collaborative team member who documents work carefully and adapts quickly to new toolchains. Seeking an embedded software engineering role.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, TypeScript, HTML/CSS, SQL

Embedded & Systems: Raspberry Pi, ARM-based SBCs, I2C, camera (CSI/OV5647), time-of-flight (VL53L7CX) sensor integration, Linux, Bash, systemd

Tools & Frameworks: Git, GCC, Flask, Node.js/Express, PostgreSQL, SQLite, ChromaDB, Ollama, OpenGL/GLSL, Pyodide, SciPy, NumPy, Pandas

Concepts: RTOS fundamentals, device drivers, interrupts, memory management, multithreading, computer architecture, digital systems, circuit analysis

Soft Skills: Clear written/verbal communication, cross-functional collaboration, mentoring, adaptability, attention to detail

CERTIFICATIONS

Intro to Software Engineering — IBM / Coursera (ID: GQFZQ443TK3X)
CI/CD for Software Developers — Codio / Coursera (ID: PM32F40ROH3B)
Software Engineer — HackerRank (ID: 7B96F29A8F9B)
Python (Basic) — HackerRank (ID: 16D74456FC3F)

EDUCATION

B.S. Electrical & Computer Engineering Technology — Expected 2027
Valencia College — Orlando, FL · Concentration: Computer Systems Technology

Associate in Arts, Physics — Graduated 2025
Valencia College — Orlando, FL

EXPERIENCE

Freelance Web Developer — Sep 2026 — Present
Front Line Refinement — remote (paid contract)

- Communicated directly with a non-technical client to translate business goals into clear technical requirements.
- Built a production business website on Node.js/Express with a PostgreSQL (Neon) data layer and a custom content-management admin panel.
- Implemented online liability-waiver collection and an appointment-booking calendar with conflict detection.
- Hardened the admin surface with bcrypt password hashing, login rate-limiting, CSRF protection, HTML sanitization, and Helmet headers.

Assistant Researcher — 2025 — 2026
Rings Research Group — Orlando, FL

- Analyzed NASA-produced astronomical datasets with Python, NumPy, and Matplotlib to produce presentation-ready results for a research team.
- Processed stellar occultation data to support binary star analysis, coordinating findings with colleagues.

- Designed a low-loss workflow for unpacking and repackaging averaged data arrays, documenting the process for reuse.

PROJECTS

Eigen — Autonomous Local AI System *Python · Ollama · ChromaDB · SQLite · Raspberry Pi*

- Designed an autonomous cognitive loop running on Linux server hardware with a layered memory system (episodic SQLite + semantic ChromaDB embeddings) and a sliding-window context manager to prevent VRAM contention.
- Integrated a Raspberry Pi 5 sensing node with an OV5647 CSI camera and an 8×8 VL53L7CX time-of-flight presence sensor for event-driven multimodal input.
- Implemented native tool-calling with a sandboxed toolset, plus output filters and strike-based IP auto-ban for safe public deployment.

Atom Wavefunction Simulator *C++ · OpenGL · GLSL*

- Architected a multi-file 3D quantum-mechanics visualization with real-time GPU ray-marching of 128^3 volumetric probability-density grids.
- Wrote custom GLSL fragment shaders and integrated AI-assisted generation within a native Wayland/EGL Linux environment.

Lagrangian Oscillating Systems Simulator *Python · Pyodide · SciPy*

- Derived equations of motion via the Lagrangian formalism for damped, forced, and driven oscillators; compiled the solver to WebAssembly (Pyodide) to run fully client-side.

OTHER EXPERIENCE

Physics & Mathematics Tutor Ongoing

Educate 2 Elevate — Orlando, FL

- Delivered advanced physics tutoring with calculus support, tailoring explanations to each student's pace and learning style.
- Built custom learning material and patiently adapted sessions for students managing severe social anxiety.

Infantry Section Leader 2018 — 2024

Army National Guard — River Falls, WI

- Managed tactical network equipment and electronic asset readiness during deployment operations.
- Led section-level operations under tight deadlines, including COVID-19 relief missions.
- Delivered technical and vehicle systems training for groups of up to 100 personnel, building communication and leadership skills.